

DJ Sindh Govt. Science College, Karachi

Preliminary Examination 2026

Mathematics – II

Tuesday
10-02-2026

Time allowed: 3 hours

For Science Pre-Engineering & Science General Groups

Maximum Marks: 100

Time allowed: 20 minutes

SECTION 'A' (MULTIPLE CHOICE QUESTIONS) – (M.C.Qs)

(Marks : 20)

- TE: (i). This section consists of 20 part questions and all are to be answered. Each question carries one mark.
(ii). The correct answer bubble must be filled on OMR sheet (I) ☐ A ☐ B ☐ C ☐ D pasted in answer script.
(iii). Use only blue / black ball point pen or pointer and avoid using pencil / White-o pen on OMR sheet.
(iv). All notations are used in their usual meanings. The use of Scientific Calculator is allowed.

1. Choose the correct answer for each from the given options.

- (1). The Maple command $\text{diff}(5 * x * y, x\$2)$ means
 - (a). *differentiates w.r.t 'x' once*
 - (b). *differentiates w.r.t 'x' twice*
 - (c). *differentiates w.r.t 'x' thrice*
 - (d). *differentiates w.r.t 'x' quare*
- (2). The output of the Maple command $\text{minimize}(\exp(x), x = 0..10)$ is:
 - (a). -1
 - (b). 0
 - (c). 1
 - (d). 10
- (3). The limit of the function $y = \frac{5x}{x+1}$ for $x \rightarrow \infty$ is _____.
 - (a). 5
 - (b). -5
 - (c). $\frac{1}{5}$
 - (d). $-\frac{1}{5}$
- (4). If $f(x) = 2x$ and $g(x) = x + 1$, then $f \circ g(-5) =$ _____.
 - (a). -1
 - (b). 1
 - (c). -8
 - (d). 8
- (5). If $y = \ln(x^2 + 4)$ then $\frac{dy}{dx} =$ _____.
 - (a). $\frac{2x}{x^2+4}$
 - (b). $\frac{2}{x^2+4}$
 - (c). $\frac{x^2+4}{2x}$
 - (d). $\frac{x^2+4}{2}$
- (6). Let f be the real valued function, defined by $f(x) = \ln e^x$, then $f^{-1}(x) =$ _____.
 - (a). $\ln e^x$
 - (b). $\ln x$
 - (c). $\ln e$
 - (d). x
- (7). $x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \dots$ is the Maclaurin's series of _____.
 - (a). $\cos x$
 - (b). $\sin x$
 - (c). $\tan x$
 - (d). $\sec x$
- (8). The point of curve which is neither point of minima nor maxima is known as point of _____.
 - (a). failure
 - (b). inflection
 - (c). stability
 - (d). middle
- (9). The domain of $\vec{r} = t^2 \hat{i} + \frac{1}{t} \hat{j} + (t+3) \hat{k}$ is:
 - (a). $\{0\} - \mathbb{R}$
 - (b). $\{0\} - \mathbb{Q}$
 - (c). $\mathbb{R} - \{0\}$
 - (d). $\mathbb{Q} - \{0\}$
- (10). $\int x^2 \ln e^{x^2} dx =$ _____.
 - (a). $\ln e^{x^2} + c$
 - (b). $\frac{x^2}{2} + c$
 - (c). $\frac{x^4}{4} + c$
 - (d). $\frac{x^3}{5} + c$
- (11). The suitable trigonometric substitution for $\int \frac{dx}{x\sqrt{x^2-3}}$ is:
 - (a). $x = 3 \sec \theta$
 - (b). $x = \sqrt{3} \sec \theta$
 - (c). $x = 3 \sin \theta$
 - (d). $x = \sqrt{3} \sin \theta$
- (12). The equation of a straight line passes through one point (x_1, y_1) and the origin, is
 - (a). $y = \frac{y_1}{x_1} x$
 - (b). $y = \frac{x_1}{y_1} x$
 - (c). $y = \frac{y_1}{x_1} + x$
 - (d). $y = \frac{x_1}{y_1} + x$
- (13). The point $P(x_1, y_1)$ lies below the line $ax + by + c = 0$, if $x_1, y_1 \in \mathbb{R}$ and $ax_1 + by_1 + c < 0$ with _____.
 - (a). $b = 0$
 - (b). $b < 0$
 - (c). $b > 0$
 - (d). $b = \pm 1$
- (14). If the plane intersects both nappes of the right circular cone but does not contain the vertex, the resulting intersection is a / an:
 - (a). parabola
 - (b). circle
 - (c). ellipse
 - (d). hyperbola
- (15). The equation of tangent to the circle $x^2 + y^2 = r^2$ at (x_1, y_1) is:
 - (a). $xx_1 + yy_1 = r$
 - (b). $xx_1 + yy_1 = r^2$
 - (c). $xy_1 + x_1y = r$
 - (d). $xy_1 + x_1y = r^2$
- (16). The slope of normal to the parabola $x^2 = 8y$ at $(4, 2)$ is:
 - (a). 0
 - (b). -1
 - (c). 1
 - (d). ± 1
- (17). The order and degree of the differential equation $\left(\frac{dy}{dx}\right)^3 = \sqrt{1 + \frac{dy^2}{dx^2}}$ are respectively:
 - (a). 1 and 2
 - (b). 2 and 1
 - (c). 3 and 2
 - (d). 2 and 3
- (18). The general solution of $xdy - ydx = 0$ is:
 - (a). $y = cx$
 - (b). $y^2 = cx$
 - (c). $y = cx^2$
 - (d). $y^2 = cx^2$
- (19). Let $f(x, y)$ and $g(x, y)$ are homogeneous functions of degrees 3 and 2 respectively, then the degree of $\frac{f(x, y)}{g(x, y)}$ is
 - (a). -1
 - (b). 0
 - (c). 1
 - (d). 12
- (20). In Simpson 1/3 method, the number of subinterval is the multiple:
 - (a). 4
 - (b). 1
 - (c). 2
 - (d). 3

ڈی۔ جے۔ سندھ گورنمنٹ سائنس کالج، کراچی

ابتدائی (پریشیم) امتحان ۲۰۲۶

روزانہ امتحان (سائنس کے تمام گروپوں کے لیے)

حصہ اول (اکثر انتخابی سوالات) ۲۰۱۵ تا ۲۰۱۷



کل نمبر: ۱۰۰

وقت: ۳ گھنٹے
۳-۲-۲۰۲۶

مندرجہ ذیل میں سے ہر ایک کے لیے دو سوالات کا جواب دیجئے۔

سوال نمبر ۱:

۱. "آدھی کا آدھا شہابی کہلی کم ہو"۔ یہ مندرجہ ذیل نصاب اس سائنس سے لیا گیا ہے:
 - (a) سر اقبال مرحوم (b) مولانا صہبہ موہانی (c) جیسے کرنا قیامت۔۔۔۔۔ (d) سحر ہونے تک
۲. ذیل نصاب "سین" "میرا الم" میں کا تحریر کردہ ہے:
 - (a) دیر آؤ (b) سڑا جالی (c) لاٹا میں صہبہ (d) مولوی صاحب
۳. انجمن سنی زبان کا مندرجہ ذیل شعر کس جگہ ہے:
 - (a) شمع لپڑ (b) غلام بابا آؤ (c) صاف چمکے ایک (d) فہم لطیف
۴. "پہاڑ جیسے 'میر' 'نظم' 'دہن' ان کی تحریر کردہ تصانیف ہیں:
 - (a) مشتاق احمد واسطی (b) پطرس بھڑی (c) مولوی صاحب (d) شبلی نعمانی
۵. 'سرف' کے معنی ہیں:
 - (a) محنت (b) دولت (c) بھڑی (d) عیبی
۶. ذیل نصاب 'ور' کی خط میں "حجم" "پاس" کا مطلب ہے:
 - (a) بہت اُسیریں (b) بہت تلاطمی (c) تنہائی کی کثرت (d) چاندی کی کثرت
۷. حمد علی شاعر کی محنتی کا گھٹا ہے:
 - (a) زور (b) اسلوب (c) طرز (d) شعرا
۸. ذیل نصاب "گیت" "ان کا تحریر کردہ ہے:
 - (a) وضاحت نسیم (b) رسا پستانی (c) نگار مسبانی (d) قاطر حسن
۹. "پینٹ کر سایہ بھی میں ناصر" شعر مکمل کیجئے:
 - (a) جب وہ رخصت ہوا، تب یاد آیا (b) ہم بہت دے دے "جب یاد آیا (c) بھر کوئی شہر طرب یاد آیا (d) تو مصیبت میں جب یاد آیا
۱۰. "رگو پتی سائے" نے اس قہمی نام سے شاعری کی ہے:
 - (a) جگر مراد آبادی (b) فراق گورکھ پوری (c) نگار مسبانی (d) رسا پستانی
۱۱. "یہ کاہنی برتن بن جانے گی" اس نثر میں اشارہ ہے:
 - (a) ایک جہود شہر بن جانے کا (b) صفائی ستھرائی کی طرف (c) لوگوں میں ذہنی استحکام پیدا کرنے کا (d) جرمی کے ایک شہر کی طرف
۱۲. ذیل نصاب "سین" "قحط الزماں" اس کتاب سے ماخوذ ہے:
 - (a) سفر نصیب (b) آواز دوست (c) نیوٹن خیال (d) چند ہم عصر
۱۳. الفاظ "شان و شوکت، عیش و عشرت، جاہ و جلال" میں ترکیب ہے:
 - (a) تضاد (b) عطفی (c) وصلی (d) اضافی
۱۴. مجاہد "خاک اڑانا" کا مطلب ہے:
 - (a) گرد اڑانا (b) دولت لانا (c) غلط لوگوں میں بیٹنا (d) بنام کرنا
۱۵. "کچ بانیے گراں مایہ" سے یہ سبق نصاب میں شامل ہے:
 - (a) میرا الم (b) سر اقبال مرحوم (c) خوش طبعی (d) مولانا صہبہ موہانی
۱۶. ذیل نصاب نعت لفظ "اعل" کا مطلب ہے:
 - (a) جمالت (b) مقرر گھڑی (c) تفا (d) پائی
۱۷. ذیل نصاب نظم "مرہم" کے شاعر ہیں:
 - (a) انور مسعود (b) حمد علی شاعر (c) رسا پستانی (d) سحر کبری
۱۸. مومن خان مومن پیشے کے اعتبار سے تھے:
 - (a) سپاہی (b) درزی (c) مزدور (d) حکم
۱۹. بالکل میں مصرعوں کی تعداد ہوتی ہے:
 - (a) ۱۱ (b) ۱۲ (c) ۱۳ (d) ۱۴
۲۰. "سر میں سدا بھی نہیں دل میں توتا بھی نہیں" اس مصرعے میں کیفیت پائی جاتی ہے:
 - (a) مثبت (b) رہائی (c) منگی (d) اضطرابی

ڈی۔ جے۔ سندھ گورنمنٹ سائنس کالج، کراچی



ابتدائی (پرلیم) امتحان ۲۰۲۶

اردو لازمی بارہویں - (سائنس کے تمام گروپس کے لیے)

نکات: ۱۰۰

ہفتہ
۱۴-۲-۲۰۲۶
وقت: ۳ گھنٹے

حصہ "ب" (مختصر جواب کے سوالات) نکات: ۴۰

نمبر 2: درج ذیل میں سے کسی دو اشعار کی تشریح شاعر کے خیال اور مختصر تعارف کے ساتھ کیجیے۔

استحقاق کیجیے مرا جب تک شوق نہ آتا نہیں ہوتا

یاد رہا! فائدہ مجھ کو ملتا ہے کس لیے لوح جہاں پہ خوب مکر نہیں ہوں میں

میں ترک نہ دوں جہوں کر ہی چکا تھا کہیں گئی اپنے میں تری لغزش پایاد؟

دل سے رخصت ہوئی کوئی خواہش مگر کہہ بے سبب نہیں آتا

تجلی کو جو پاں جلوہ نہ دیکھا ہمارے دنیا کو دیکھا نہ دیکھا

نمبر 3: (الف) منتخب کردہ اقتباس کے سین کا نام اور مصنف کا نام تحریر کیجیے۔

(ب) درج ذیل میں سے کسی ایک اقتباس کی تشریح کیجیے۔

۱) جھوٹ کی تعمیل ہر وقت بھری ہے، کبھی خالی نہیں ہوتی۔ خیر ہے، ایسا کم سخت ہے کہ جو ہاتھ اسے لٹا دیتا ہے۔

اسی کو کاٹ کھاتا ہے اور دوست، دشمن دونوں کی ہمارے خاک اڑاتا ہے۔ سبب اس کا یہ ہے کہ انسانیت سے خارج ہے۔

۲) ایک حاکم کے لیے محکوم کی زندگی، ایک محکوم کے لیے حاکم کی زندگی، ایک دولت مند کے لیے غریب کی زندگی اور ایک غریب کے لیے دولت مند کی زندگی

کامل مثال نمونہ نہیں بن سکتی۔ اس لیے ضروری ہے کہ عالم گیر اور دینی مضامین کی زندگی ان تمام مختلف مناظر کے رنگ پر رنگ پھولوں کا گلہ دستہ ہو۔

۳) معجزے کا فائدہ نہیں ہوا، لیکن محبت و خلوص میں اب بھی بڑی کراستیں پوشیدہ ہیں۔

نمبر 4: (الف) منتخب کردہ نظم کے شاعر کا مختصر تعارف تحریر کیجیے۔

(ب) درج ذیل میں سے ایک نظم کا مرکزی خیال تحریر کیجیے۔

۱) ہلی کا چاند
۲) حضرت فاطمہ الزہراء کی رخصتی
۳) چاند اور تارے

(الف) درج ذیل جملے کے شاعر اور نظم کا نام تحریر کیجیے۔

(ب) درج ذیل جملے کی تشریح کیجیے۔

۱) سچا فکر میں ہے تائبہ عرض بریں خاک سے پائے مقدس کا لگا کر منزل

آرزو ہے کہ وہ دھیمان ترا تادم مرگ فکل تیری نظر آنے مجھے جب آنے اہل

نمبر 5: (الف) منتخب کردہ سین کے مصنف کا مختصر تعارف تحریر کیجیے۔

(ب) مندرجہ ذیل میں سے کسی ایک سین کا خلاصہ تحریر کیجیے۔

۱) سحر ہونے تک
۲) سورے ۷۰ کل آنکھ میری کھلی
۳) زمین

مندرجہ ذیل سوالات میں سے کسی پانچ کے جوابات تحریر کیجیے۔

۱۔ عالم گیر اپنے جینے بیٹے مرنا کام بخش کو "کوک" کے ساتھ قید کرنے کا حکم کیوں دیا؟

۲۔ "اونٹ کو گھنٹی پہننے" کا مطلب ہے؟

۳۔ انسان اپنے ماضی کو اس قدر عزیز کیوں رکھتا ہے؟

۴۔ کالونی کے ادبی نڈن اور سیاسی نڈن سے کیا مراد ہے؟

۵۔ طرہی مشاعرہ کن شعرا کی یادگار ہے؟

۶۔ دلنری پے کی کہانی سے کیا سبق ملتا ہے؟

۷۔ اچھے کھیل کی کیا خوبی ہے؟

حصہ "ج" (تفصیلی جواب کے سوالات) نکات: ۶۰

نمبر 6: مندرجہ ذیل میں سے کسی ایک شاعر کے کلام کی شاعرانہ خصوصیات تحریر کیجیے۔

۱) مرزا اسد اللہ خان غالب
۲) علامہ اقبال
۳) میر تقی میر

نمبر 7: مندرجہ ذیل میں سے کسی ایک نثر نگار کے طرزِ تحریر کی خصوصیات تحریر کیجیے۔

۱) مولوی مہاگن
۲) سید سلیمان ندوی
۳) پطرس بخاری

دی گئی عبارت کو پڑھ کر سوالات کے جوابات لکھیے:

"خدا آباد رکے، کراچی کا کیا کتنا! کوئی شخص راہ چلتے ہیں ی پان کی پیک تھوک دے اور ہر اس کی طرف ٹکٹنگی بازو کر دیکھنے لگے تو دو منٹ میں ٹکٹ کے ٹکٹ لگے جائیں اور سارا ٹریک رُک جائے۔ یاد کرو! آٹا سے میں ہاں تماشائی کی تیلی سے پڑتی ہے۔ نہ کہ ماری کی ڈھنگی سے! مرنا لے بات کو کسا سے کسا کہ ہٹا ہٹا۔"

۱۔ اس عبارت کا عنوان تجویز کیجیے۔

۲۔ "ٹکٹنگی ہڈنے" سے کیا مراد ہے؟

۳۔ "خدا آباد رکے" یہ جملہ مصنف نے کس کے لیے کہا ہے اور کیوں کہا؟

۴۔ عبارت کو پڑھ کر تلخیص کیجیے۔

۵۔ عبارت میں ہاں کس سے پڑتی ہے اور کیوں؟

SECTION 'A'**(MULTIPLE CHOICE QUESTIONS) (M.C.Qs) Marks 10**

- Q. 1 Choose the correct answer for each from the given option:
- This province was separated from Bombay in 1936 following Muslim demands:
(a). Punjab (b). NWFP (c). Sindh (d). Balochistan
 - The Lower House of parliament in Pakistan is called:
(a). Senate (b). National Assembly
(c). House of Commons (d). Congress
 - The longest river in Pakistan is:
(a). Ravi (b). Jhelum (c). Chanab (d). Indus
 - The Khilafat Movement (1919–1924) was launched to protect the status of:
(a). Al-Aqsa Mosque (b). Ottoman Caliphate
(c). Mughal Empire (d). holy places of Hijaz
 - He was the Prime Minister of Pakistan during the enforcement of the 1956 Constitution:
(a). Muhammad Ali Bogra (b). Chaudhry Muhammad Ali
(c). Shaheed Suhrawardy (d). Feroz Khan Noon
 - The pamphlet 'Now or Never' was written by:
(a). Chaudhry Muhammad Ali (b). Allama Iqbal
(c). M.A. Jinnah (d). Muhammad Ali Johar
 - This Sufi poet composed poetry in seven languages:
(a). Shah Abdul Latif (b). Waris Shah
(c). Bulleh Shah (d). Sachal Sar Mast
 - The first annual session of All India Muslim League was held in this city:
(a). Karachi (b). Lahore (c). Aligarh (d). Lucknow
 - The first Martial Law was imposed in Pakistan in:
(a). 1956 (b). 1958 (c). 1969 (d). 1977
 - Simla Agreement was signed between Zulfikar Ali Bhutto and Indira Gandhi in:
(a). 1970 (b). 1971 (c). 1972 (d). 1973

SECTION 'B'**(SHORT-ANSWER QUESTIONS) (20 MARKS)**

NOTE: Attempt any Five part questions. All part questions carry equal marks.

Q. 2.

- What was the vision of the Quaid-e-Azam OR Allama Iqbal regarding the ideology of Pakistan
- What were the reasons behind launching the Khilafat Movement in 1919?
- What do you know about Indus Basin treaty?
- In what ways did Islam transform the social structure of South Asia?
- Explain Indo-Pak diplomatic relations?
- Write five Islamic provisions of 1973 Constitution?
- State the significant features of Objective Resolution?
- Describe any five principles of Foreign policy of Pakistan?

SECTION 'C'**(DETAILED-ANSWER QUESTIONS) (20 Marks)**

NOTE: Answer any Two questions from this section. All questions carry equal marks.

Q. 3. What challenges did Pakistan encounter immediately after independence?

Q. 4. Describe the role of Judiciary in protecting fundamental rights of citizens.

OR

Why was Pakistan established? of Pakistan? Discuss its aims and objectives.

Q. 5. Describe different languages spoken in Pakistan?

Good Luck

DJ SINDH GOVT. SCIENCE COLLEGE, KARACHI

Preliminary Examination 2026

Chemistry-XII

Wednesday
11-02-2026

Time allowed: 3 hours

(Science Pre-Engineering & Pre Medical Group)

Maximum Marks: 85

Time allowed: 20 minutes

SECTION 'A' (MULTIPLE CHOICE QUESTIONS) – (M.C.Q.s) (Marks: 17)

- NOTE: i) This section consists of 17 part questions and all are to be answered. Each part question carries one mark.
ii) Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part.
iii) The use of scientific calculator is allowed

Q. 1. Choose the correct answer for each from the given options.

(i) The oxidation states of the elements of group VII A:

- * +1 and +3 * +1, +5 and +5 * +1, +1, +3, +5 and +7 * +3, +3 and +5

(ii) The metal is used in making jewelry and also serves as catalyst:

- * Pt * Fe * Pb * Al

(iii) Lucas reagent is a mixture of:

- * Zn and Hg * Zn and HCl * $ZnCl_2$ and HCl * NaOH and CaO

(iv) The best oxidizing agent among halogen is:

- * I_2 * Br_2 * Cl_2 * F_2

(v) Which one of the following does not belong to alkaline earth metal?

- * Be * Ba * Ni * Ca

(vi) This is not a nucleophile:

- * NH_3 * OH^- * CN^- * NH_4^+

(vii) Welding gas among the following is:

- * Ethene * Ethane * Methane * Ethyne

(viii) The element forms peroxide with oxygen:

- * Na * K * Li * Mg

(ix) Formic acid is naturally found in:

- * Venom of ants * Bee sting * Vinegar * Butter

(x) Saponification is the formation of soap by the reaction fat and oil with:

- * Sugar * HCl * An Alkali * Glycerol

(xi) The coordination number of Ni in the complex compound $[Ni(en)_2(NO_2)_2]^{+2}$:

- * 2 * 4 * 6 * 5

(xii) Welding gas among the following is:

- * Methane * Ethane * Ethylene * Acetylene

(xiii) Benzene burns with smokey flame because of its:

- * Inflammability * High carbon percentage * High resonance energy * Aromaticity

(xiv) Drugs that lower the body temperature to normal are known as:

- * Antibiotics * Antipyretic * Antiallergic * Anti histamines

(xv) This element is used to build strong bones and teeth:

- * Zn * Ca * Fe * Mg

(xvi) Ozone is produced in the stratosphere region due to photo chemical reaction of sun rays and:

- * CFCs * NO_2 * NO * O_2

(xvii) Infra-red spectroscopy is a technique use to determine:

- * Functional group * Double and triple bonds * Mass to charge ratio * Conjugated system

DJ SINDH GOVT. SCIENCE COLLEGE, KARACHI



Preliminary Examination 2026

Chemistry – XII

Wednesday
11-02-2026

Time allowed: 2 hrs. and 40 min. (Science Pre-Engineering & Pre Medical Group)

Max Marks: 68

SECTION 'B' (SHORT-ANSWER QUESTIONS) (Marks : 36)

Note: Answer any Nine part questions. Select Four part questions from Inorganic – General Chemistry and Five part questions from Organic Chemistry. All part questions carry equal marks.

INORGANIC-GENERAL CHEMISTRY

(Q.2) (i) Define the flame test. Briefly explain why alkali and alkaline earth metals produce colored flames in this test. State the flame colors for any two alkali metals. OR Give reasons for any four of the following:

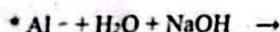
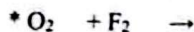
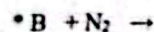
* Boiling point of halogens increases down the group in the periodic table. * Electronegativity decreases regularly from top to bottom in s-block elements. * Alkali metals are good conductor of electricity. * Fluorine is the strongest oxidizing agent. * Multidentate ligands are known as chelating agents.

(ii) Write the diagonal relationship between lithium and magnesium. OR Write the three properties of beryllium that show its unique behavior in group II A. Also write the reason why beryllium does not react with cold water and steam?
(iii) Define d-block elements, why do they form colored compounds? Explain it in term of Crystal Field Theory.

(iv) Write IUPAC names of the following complexes:

* $[Cu(NH_3)_4]SO_4$ * $[Fe(CN)_5NO]^{2-}$ * $K_3[Cr(C_2O_4)_2Cl_2]$ * $[Pt(en)_2(NO_2)_2]Cl_2$

(v) Describe the preparation and two properties of Nylon and Polyvinyl chloride. OR Complete the following reactions.



(vi) Define green house gases. How green house gases cause to global warming? OR What are carbohydrates? Give their classification on the basis of structure.

(vii) What information about the structure of a molecule we can get from mass spectroscopy? Give the applications of mass spectroscopy.

ORGANIC CHEMISTRY

(viii) Define Bucky ball. Explain its structure and write two properties. OR Define chiral carbon and plane polarized light. Explain optical isomerism briefly. OR Define Isomerism. Explain different types of Structural isomerism with examples.

(ix) Two organic compounds A and B are given below: (A) $CH_2=CH_2$ (B) $CH \equiv CH$

* Draw the orbital structure of compound A and specify: * Hybridization * Bond angles * Write the ozonolysis reactions of A and B under suitable conditions. OR Give the equations and write the name of the final product in the following process.

* Ethyne is treated with hydrogen bromide. * Ethene is treated with oxygen in presence of peracetic acid at $100^\circ C$.

* 1, 2-dibromoethane is heated with alcoholic KOH. * Ethyne is treated with H_2O in the presence of $H_2SO_4/HgSO_4$.

(x) Define organometallic compounds. Give two examples of organo metallic compounds. Write the equation of methyl magnesium iodide with: * CH_3CHO * CO_2

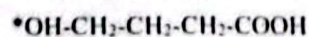
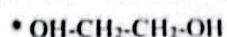
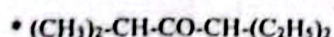
(xi) Why are amines basic in nature? How can we prepare ethyl amine from the following compounds? * ethyl iodide * methyl cyanide * ethanamide OR What is photochemical reaction? Write the mechanism of chlorination of methane.

(xii) Name four derivative of carboxylic acid and write the equations of their preparation. OR Define Homologous series and write its three general properties.

(xiii) Identify each of the following with one laboratory test: * Alcohol * Phenol * Alkene * Aldehyde

OR Explain electrophilic substitution reaction mechanism in benzene with reference to halogenations.

(xiv) Write the IUPAC names of any four of the following:



SECTION 'C' (DETAILED-ANSWER QUESTIONS) (Marks : 32)

Note: Answer any **Four** questions. Two question from Inorganic – General Chemistry and Two from Organic Chemistry. All questions carry equal marks.

INORGANIC-GENERAL CHEMISTRY

Q.3. Explain with the help of diagram of Castner Kellner cell, how caustic soda is obtained by electrolysis of aqueous sodium chloride. Write two advantages and disadvantages of the Castner Kellner process. **OR** Describe the manufacturing of H_2SO_4 by contact process. Show that by chemical equation it acts as oxidizing and dehydrating agent.

Q.4. Describe various steps involve in the extraction of 99.99% pure copper from its chalcopyrite ore. (Draw diagram where necessary) **OR** Write the balance chemical equations for the following:

- * A mixture of carbon and silicon is heated under elevated temperature. * Phosphorus is put into water.
- * Bleaching powder is treated with hydrochloric acid. * Chlorine gas is passed through hot aqueous solution of caustic soda. * Copper is treated with concentration nitric acid. * A piece of chromium is put into dilute hydrochloric acid.
- * Reaction between KMnO_4 and FeSO_4 in the presence of H_2SO_4 (write ionic equation)
- * Reaction between $\text{K}_2\text{Cr}_2\text{O}_7$ and FeSO_4 in the presence of H_2SO_4 (write ionic equation)

Q.5. Define Troposphere and stratosphere. Describe the chemistry involved due to the presence of oxides of carbon and nitrogen in the troposphere. (Write equations where necessary)

ORGANIC CHEMISTRY

Q.6. What is nucleophilic substitution reaction? Explain S_N^1 and S_N^2 mechanism with examples. **OR** Define β – elimination reaction. Discuss the mechanism of E_1 and E_2 reactions with examples. **OR** Define protein. Discuss types of protein on the basis of their function (any three). Write two properties and two biological importance of protein.

Q.7. Draw and explain the molecular orbital structure of Benzene. **OR** What is meant by orientation of benzene? Explain ortho-para and meta directing group. Write the equation for the preparation of TNT and m-nitro toluene from benzene.

Q.8. Draw the structure of the following organic molecules:

- | | | | |
|-----------------------|--------------------------|--------------------|---------------------------|
| * Isopropyl butanoate | * Ethyl neo-pentyl ether | * Divinylacetylene | * p-cresol |
| * Pyrogallol | * Neo Pentyl iodide | * Benzamide | * Benzene-1, 4-dioic acid |

OR Write the equations for the following reactions:

- * Oxidation of 2° -alcohol with $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$
- * Reaction of phenol with H_2SO_4 at 20°C
- * Reaction of propanone with hydroxyl amine.
- * Ethanol, in excess, is heated up to 140°C in presence of H_2SO_4
- * Dehydration of ethyl alcohol at 170°C in conc. H_2SO_4
- * Reaction of ethylene glycol with periodic acid.
- * Reduction of acetone with LiAlH_4
- * Oxidation of 1° -alcohol with PCC.

Good Luck

DJ SINDH GOVT. SCIENCE COLLEGE, KARACHI

Preliminary Examination 2026

Physics – XII

Thursday
12-02-2026

Time allowed: 3 hours

For Science (P.E.), Science (P.M.) & Science (Gen.) Groups

Maximum Marks: 85

Time allowed: 20 minutes

SECTION 'A' (MULTIPLE CHOICE QUESTIONS) – (M.C.Qs.)

(Marks : 17)

NOTE: (i). This section consists of 17 part questions and all are to be answered. Each question carries one mark.

(ii). The correct answer bubble must be filled on OMR sheet (I) (A) (B) (C) (D) pasted in answer script.

(iii). Use only blue / black ball point pen or pointer and avoid using pencil / White-o pen on OMR sheet.

(iv). All notations are used in their usual meanings. The use of Scientific Calculator is allowed.

Q. 1. Choose the correct answer for each from the given options.

- (i). In kinetic theory of gases, a molecule of mass m of an ideal gas collides with a wall of vessel with velocity v . The magnitude of change in linear momentum of the molecule is
(a). $-2mv$ (b). $-mv$ (c). mv (d). $2mv$
- (ii). As $C_p - C_v = R$ shows that $C_p > C_v$, what is also true?
(a). $\Delta T_p > \Delta T_v$ (b). $\Delta U_p > \Delta U_v$ (c). $\Delta U_p = \Delta U_v$ (d). $\Delta U_p < \Delta U_v$
- (iii). The second law of thermodynamics may be written in usual symbols, as
(a). $\Delta Q = \Delta u + \Delta W$ (b). $\Delta u = \Delta Q + P\Delta V$ (c). $\Delta Q = T\Delta S$ (d). $\Delta S = T\Delta Q$
- (iv). The magnetic force of deflection on a charged particle undergoing a circular motion in a uniform magnetic field is independent of its
(a). speed (b). mass
(c). charge (d). magnetic field strength
- (v). Which of the following apparatus construction uses electromagnetic induction?
(a). Galvanometer (b). Electric motor (c). voltmeter (d). AC generator
- (vi). The RMS value of an AC voltage is 220 V. What is the peak (maximum) value of the voltage?
(a). 115V (b). 220V (c). 311V (d). 440V
- (vii). What does a higher shear modulus of a material indicate?
(a). High flexibility (b). Large deformation
(c). Weak structure (d). Strong resistance to shear
- (viii). What is the net electric charge on an n-type semiconductor?
(a). positive (b). negative
(c). zero (d). depends upon temperature
- (ix). A door lock system opens only when both the correct passcode and a keycard are present. Which logic gate best represents this system?
(a). OR (b). AND (c). NAND (d). NOR
- (x). According to a moving observer, time on a stationary clock appears:
(a). Faster (b). slower (c). same (d). infinite
- (xi). The Division and Germer experiment indicate _____ of electrons.
(a). Interference (b). diffraction (c). polarization (d). refraction
- (xii). If the electron in a hydrogen atom jumps from the 3rd orbit to second orbit the emitted radiation has the wavelength
(a). $5R/36$ (b). $36/5R$ (c). $6/5R$ (d). $5/6R$
- (xiii). Half-life of radioactive element depends upon its
(a). temperature (b). nature (c). mass (d). pressure
- (xiv). What may not be conserved in nuclear process
(a). Nucleons number (b). Atomic number (c). neutrons number (d). All of these
- (xv). Which of the following is a lepton
(a). electron (b). π pion (c). meson (d). proton
- (xvi). In which device does the current lead the voltage by 90° ?
(a). resistor (b). Inductor (c). capacitor (d). transformer
- (xvii). Which of the following uses electric energy and does not convert it into any other form?
(a). bulb (b). electric motor (c). heater (d). transformer

DJ SINDH GOVT. SCIENCE COLLEGE, KARACHI

Preliminary Examination 2026

Physics – XII

Thursday
12-02-2026

Time allowed: 2 hrs. and 40 min. For Science (P.E.), Science (P.M.) & Science (Gen.) Groups Max Marks: 68

SECTION 'B' (SHORT-ANSWER QUESTIONS) (Marks : 36)

- Q. 2. Answer any **nine** question-parts from this section. All questions carry equal marks. (i. e. 4 marks of each part).
- Describe the working of a transformer.
 - A heat engine works at the rate of 500kW. The efficiency is 30%. Calculate the loss of heat per hour.
 - A wire 1.80m long carries a current of 13.0A and makes an angle of 35.0° with a uniform magnetic field of magnitude 1.50T. Calculate the magnetic force on the wire.
 - Two coils are placed close to each other. If a change in current of 3A/s in the first coil induces an EMF of 4V in the second coil. Calculate the mutual inductance.
 - Write the β^+ decay process for each of the following nuclei with their proper chemical symbols including Z and A for each daughter nucleus a) ${}^{19}_8\text{F}$ b) ${}^{15}_8\text{O}$.
 - Draw a graph of ultimate tensile stress Vs strain & differentiate elastic deformation and plastic deformation with the help of the graph drawn.
 - What is the Compton effect? Explain its significance and discuss its formula in terms of Compton shift and Compton wavelength.
 - In a parallel RC circuit, the resistance is 60Ω and the capacitance is $30\mu\text{F}$. Calculate the total current flowing through the circuit at a frequency of 120Hz.
 - At what velocity does the kinetic energy of a particle equal its rest energy?
 - Calculate the molar specific heat capacity of a gas when 2moles of the gas absorb 1500J of heat energy and its temperature increases by 25°C.
 - Choose any two logic gates symbols and explain their operation.
 - What is meant by rectification? How does a semiconductor diode work as a full wave rectifier?
 - List all types of quarks, leptons, and force carrier particles.
 - The elastic limit of copper is $1.5 \times 10^8 \text{N.m}^2$. It is to be stretched by a load of 10kg. Find the diameter of the wire if the elastic limit is not to be exceeded.

SECTION 'C' (DETAILED-ANSWER QUESTIONS) (Marks : 32)

NOTE: Attempt any **two** question from this section. Each question carries 16 marks. Draw diagrams, where necessary.

- Q.3(a). State the postulates of Bohr's atomic model. Using these postulates, derive an expression for the binding energy of an electron in the n th orbit of a hydrogen atom.
- (b). Describe the moving-coil galvanometer. Explain its construction and operation, and show how the deflection of the coil depends on the current flowing through it.
- Q.4(a). State any three postulates of kinetic molecular theory and derive the expression for gas pressure in terms of density and average square velocity.
- (b). Define natural radioactivity, state the law of radioactive decay, discuss activity, half-life and rate of decay, and derive the mathematical expression of the decay law.
- Q.5(a). Explain the laws related to electromagnetic induction. Mention its types and derive the relevant formula for emf induced in self-induction.
- (b). ~~1st~~ Carnot engine? Describe its construction and working & derive the expression for its efficiency.

OR

What is a mass spectrograph? Explain its construction, principle and working. Derive its relevant mathematical expression.

Good Luck